

Yixuan Hua

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Education

Princeton University, Princeton, NJ, United States

Aug 2022 - Present

Ph.D. in Operations Research & Financial Engineering

Advisor: Amir Ali Ahmadi, Bartolomeo Stellato

Peking University, Beijing, China

Sep 2018 - July 2022

B.S. in Mathematics, Overall GPA: 3.82/4.0, Major Rank: 5/37

Advisor (undergraduate research): Zaiwen Wen

Professional Experience

Flexcar

Boston, MA

Strategic Intern – Finance

June 2025 - Aug 2025

- Built statistical and mathematical models (logistic regression, survival analysis) to analyze customer conversion, tenure, and price sensitivity, enabling data-driven understanding of behavioral drivers.
- Developed an individualized pricing strategy using models' estimates of conversion and tenure to optimize revenue and customer lifetime value.
- Analyzed order velocity using linear regression and evaluated its relationship with inventory size and distribution.

Publications

Journal Articles

Overparameterized ReLU Neural Networks Learn the Simplest Model: Neural Isometry and Phase Transitions.

Y. Wang, **Y. Hua**, E. J. Candès and M. Pilanci.

IEEE Transactions on Information Theory, doi: 10.1109/TIT.2025.3530355.

Preprints

Disjunctive Sum of Squares.

A. A. Ahmadi, S. Dash, **Y. Hua**, B. Stellato.

arXiv e-prints:2605.28674, 2026.

Talks

"Disjunctive Sum of Squares"

INFORMS at Seattle, 2024

ICCOPT at University of Southern California, 2025

Honors and Awards

Merit Student, 2018-2019 academic year

The Suzhou Industrial Park Scholarship, 2018-2019 academic year

First Prize, The Jiang Zehan Mathematical Contest in Modeling, 2019

First Prize, The Mathematics Competition of Chinese College Students, 2019

First Prize in Zhejiang Province, National High School Mathematics League, 2016 & 2017

Silver Medal, China Mathematics Olympiad, 2016

Skills

Programming: Python, R, MATLAB